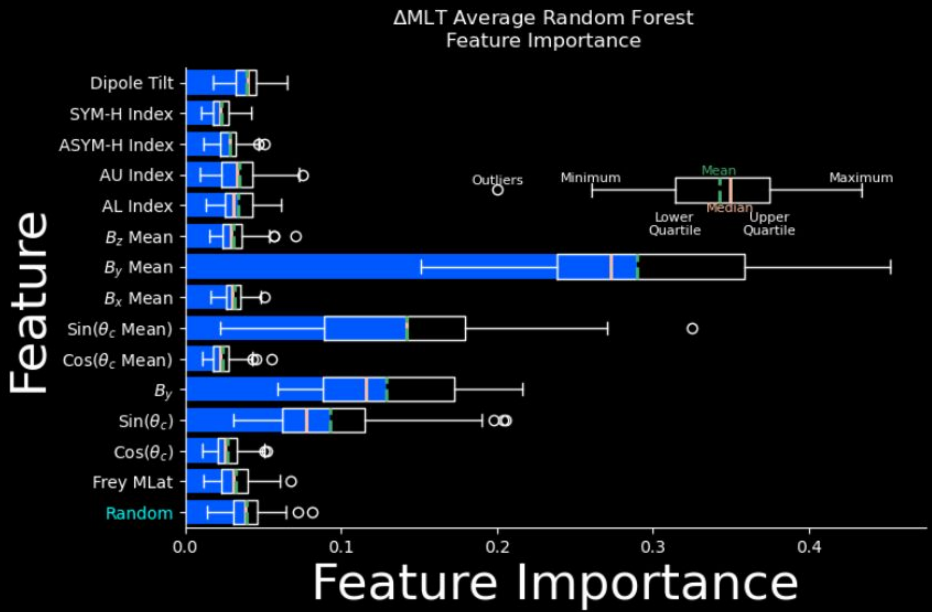
Decision Trees and Feature Importance



Decision Trees

- **Decision Tree**
- Root node
- Child node
- Left node
- Right node
- **Features**
- **Samples**
- **Target Variable**

Experiment

100 Students took a test

What we know:

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Pass/Fail**

What we want to know:

1. Which factor contributes most to whether a student passes?

2. If another student took the test, based on the prior knowledge, will they pass or fail?

Decision Trees

- **5 Features** Hrs studied, Hrs slept, Attendance Rate, Previous Exam Score, Age
- **Target Variable** Pass/Fail
- **100 Samples** Students

Decision Trees

Root Node

100 Samples

<u>Root Node</u>

- Randomly Select a subset of **Features**

Decision Trees

Hrs studied, Hrs slept, Attendance Rate, Previous Exam Score, Age

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Hrs studied, **<u>Hrs slept</u>**, **<u>Attendance Rate</u>**, Previous Exam Score, **<u>Age</u>**

Root Node

100 Samples

<u>Root Node</u>

- Randomly Select a subset of **Features**

Decision Trees

Root Node

- Randomly Select a subset of **Features**

Loop through **Hrs slept**, **Attendance Rate**, **Age**

Root Node

100 Samples

Decision Trees

Hrs slept (x)

[1, 10, 6, 3 …]

Test each unique value as a threshold

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Hrs slept (x)

[1, 10, 6, 3 …]

Threshold

$x<1$, $x \geq= 1$

Left | Right

How well split are pass and fail?

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Hrs slept (x)

[1, 10, 6, 3 …]

Threshold

x<10, x≥=10

Left | Right

How well split are pass and fail?

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Hrs slept (x)

[1, 10, 6, 3 …]

Threshold

x<6, x≥=6

Left | Right

How well split are pass and fail?

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Hrs slept (x)

[1, 10, 6, 3 …]

Threshold

x<3, x≥=3

Left | Right

How well split are pass and fail?

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Hrs slept (x)

[1, 10, 6, 3 …]

Threshold

x<..., x≥=...

Left | Right

How well split are pass and fail?

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Attendance Rate (x)

[1, 10, 6, 3 …]

Threshold

x<..., x≥=...

Left | Right

How well split are pass and fail?

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Age (x)

[1, 10, 6, 3 …]

Threshold

x<..., x≥=...

Left | Right

How well split are pass and fail?

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Trees

List of Impurity
Reduction Values

Which one is best?

Hrs Slept

Hrs Slept
4

100 Samples

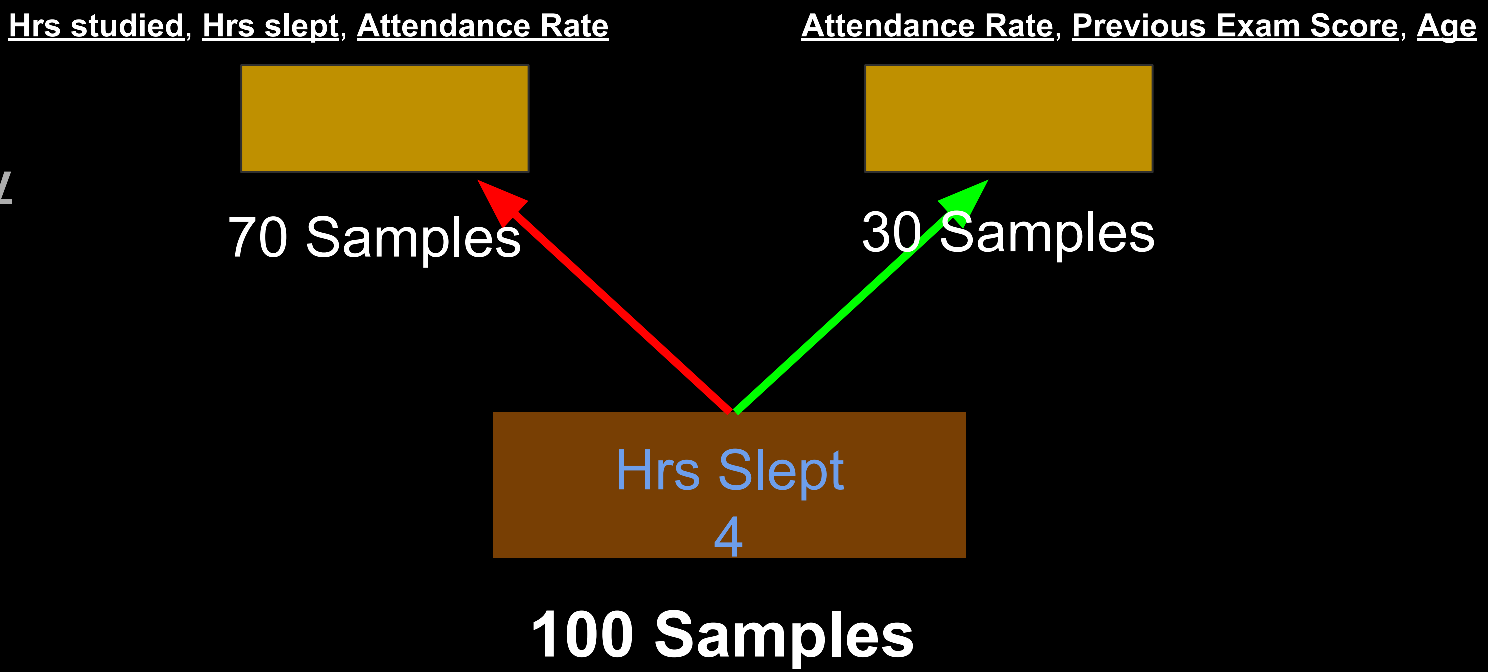
Root Node

- Randomly Select a subset of **Features**

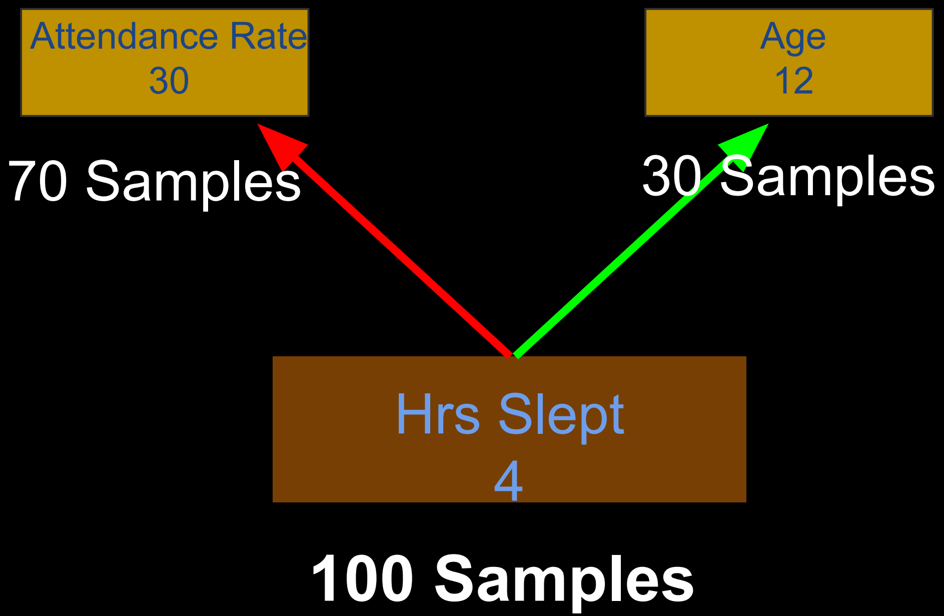
Decision Trees

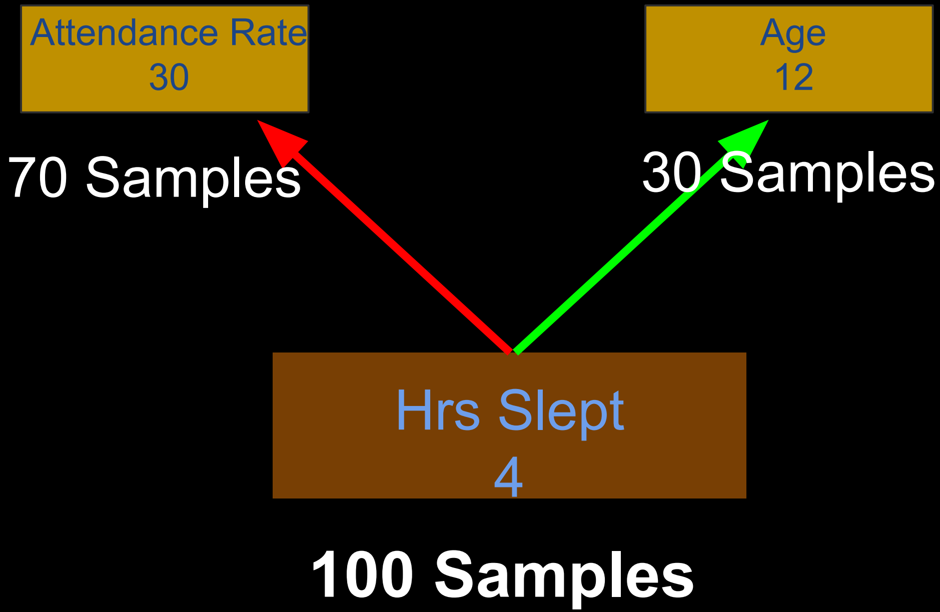
Decision Trees

Randomly Select Features

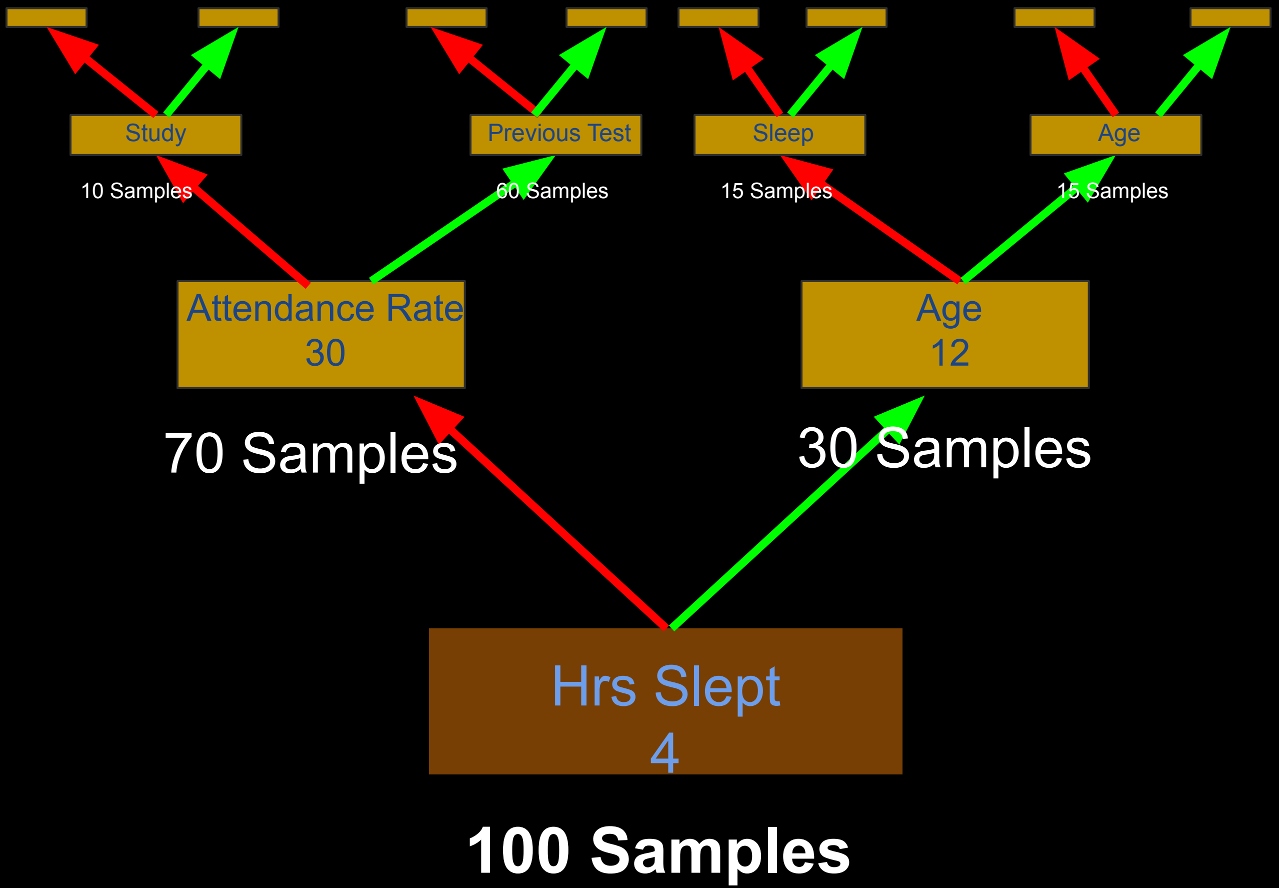


Decision Trees





Attendance Rate
30
Age
12
70 Samples
30 Samples
Hrs Slept
4
100 Samples



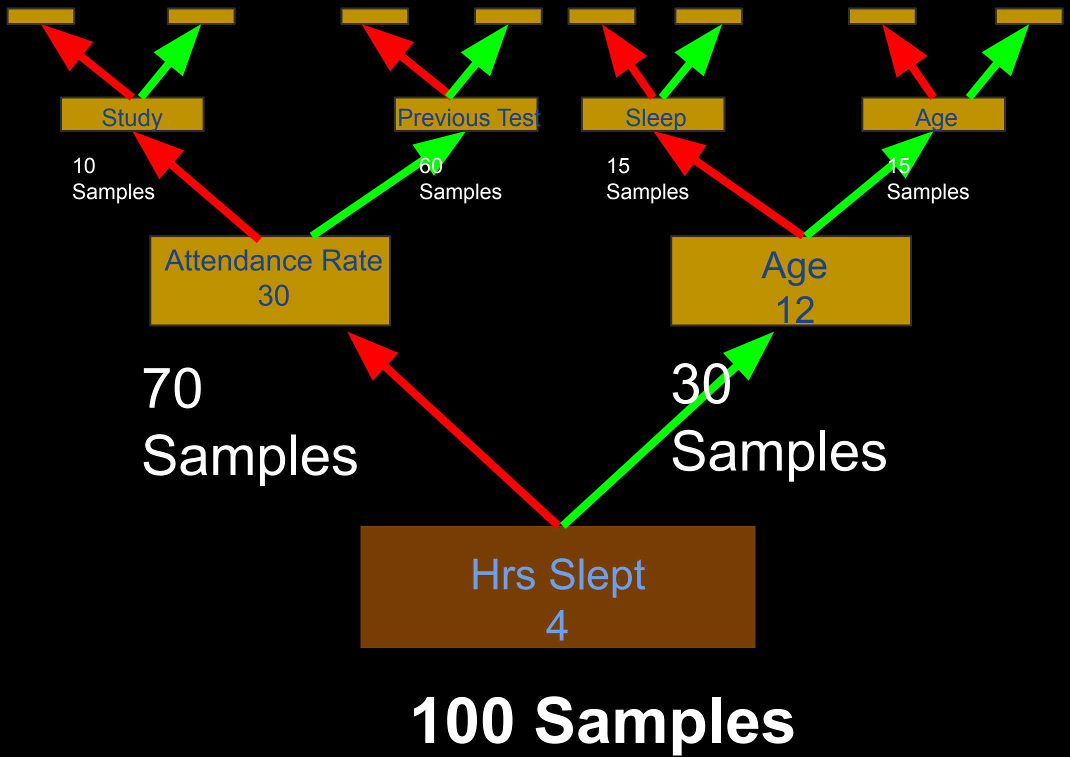
Study
Previous Test
Sleep
Age
10 Samples
60 Samples
15 Samples
15 Samples
Attendance Rate
30
Age
12
70 Samples
30 Samples
Hrs Slept
4
100 Samples

Decision Trees

- **Decision Tree**
- Root node
- Child node
- Left node
- Right node
- **Features**
- **Samples**
- **Target Variable**

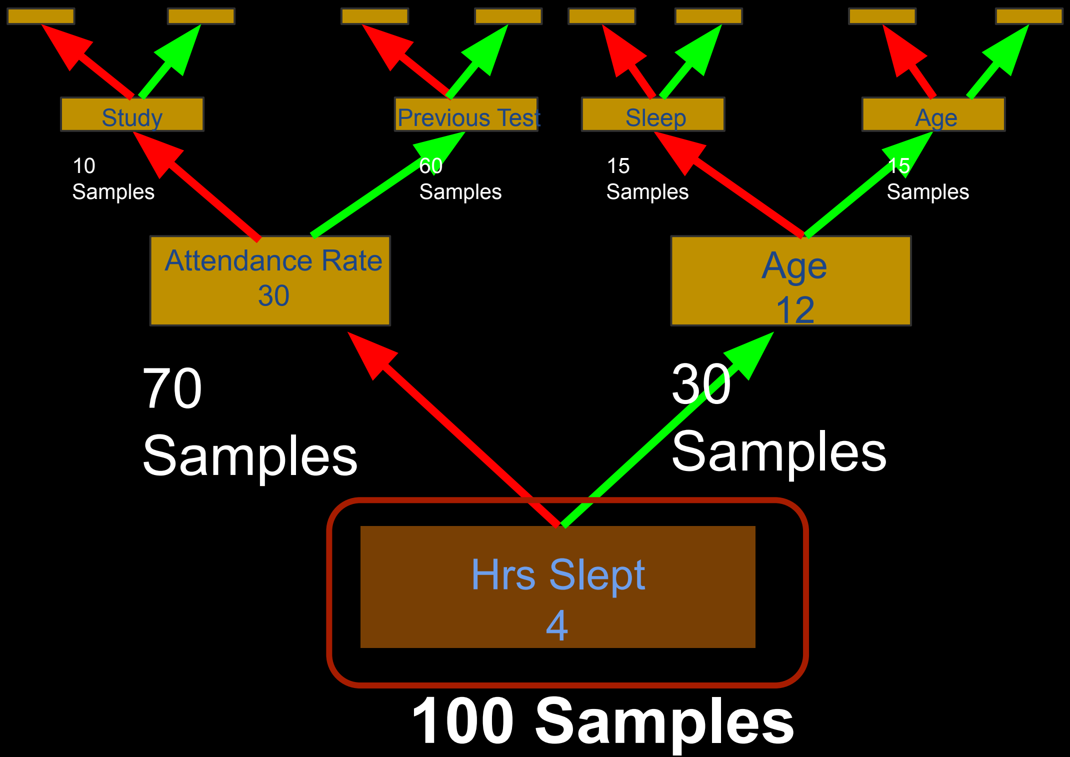
Decision Trees

- **Decision Tree**
- Root node
- Child node
- Left node
- Right node
- **Features**
- **Samples**
- **Target Variable**



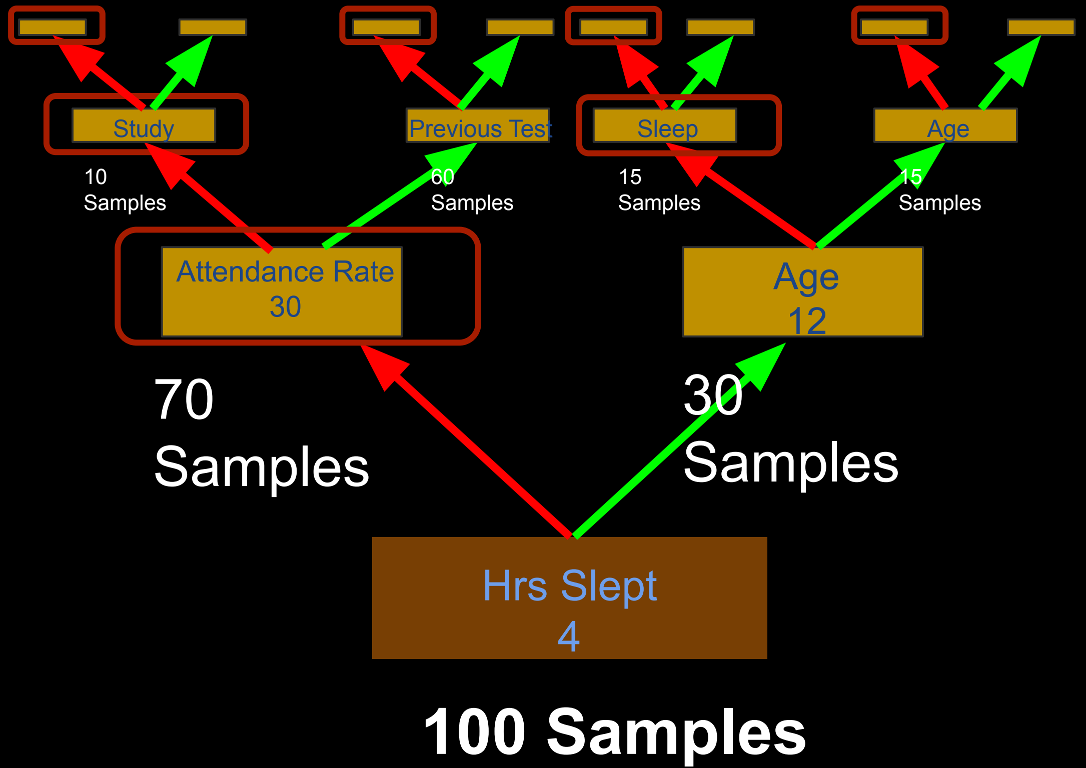
Decision Trees

- **Decision Tree**
- Root node
- Child node
- Left node
- Right node
- **Features**
- **Samples**
- **Target Variable**



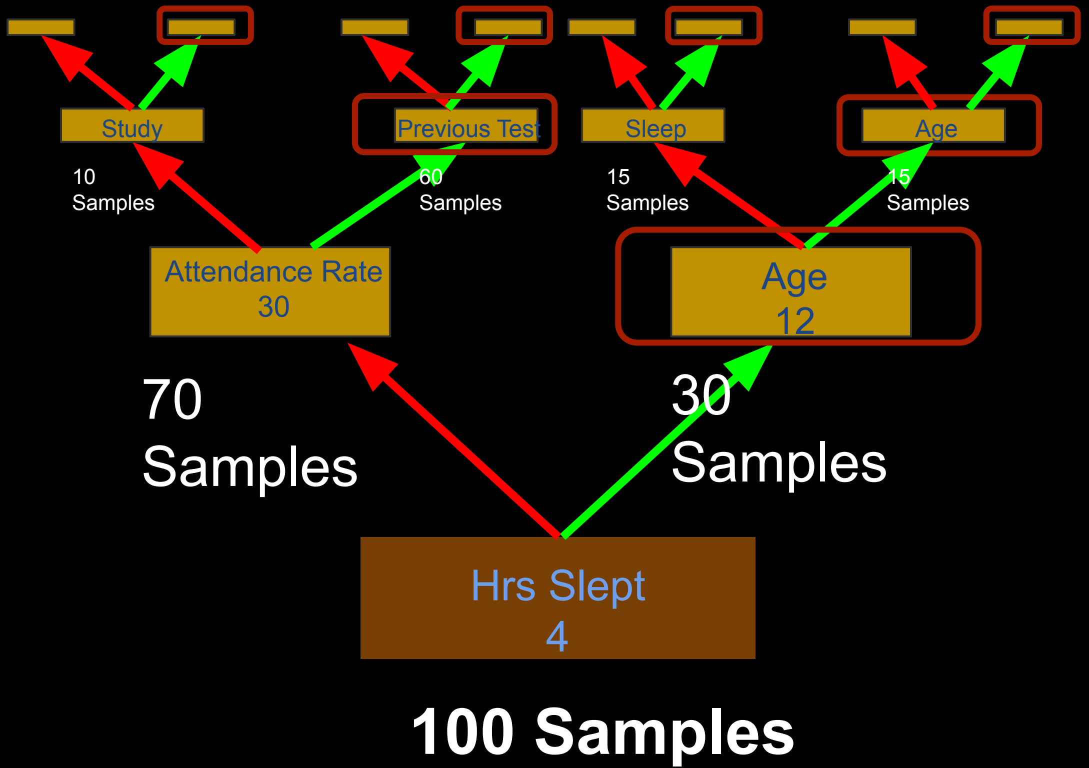
Decision Trees

- **Decision Tree**
- Root node
- Child node
- Left node
- Right node
- **Features**
- **Samples**
- **Target Variable**



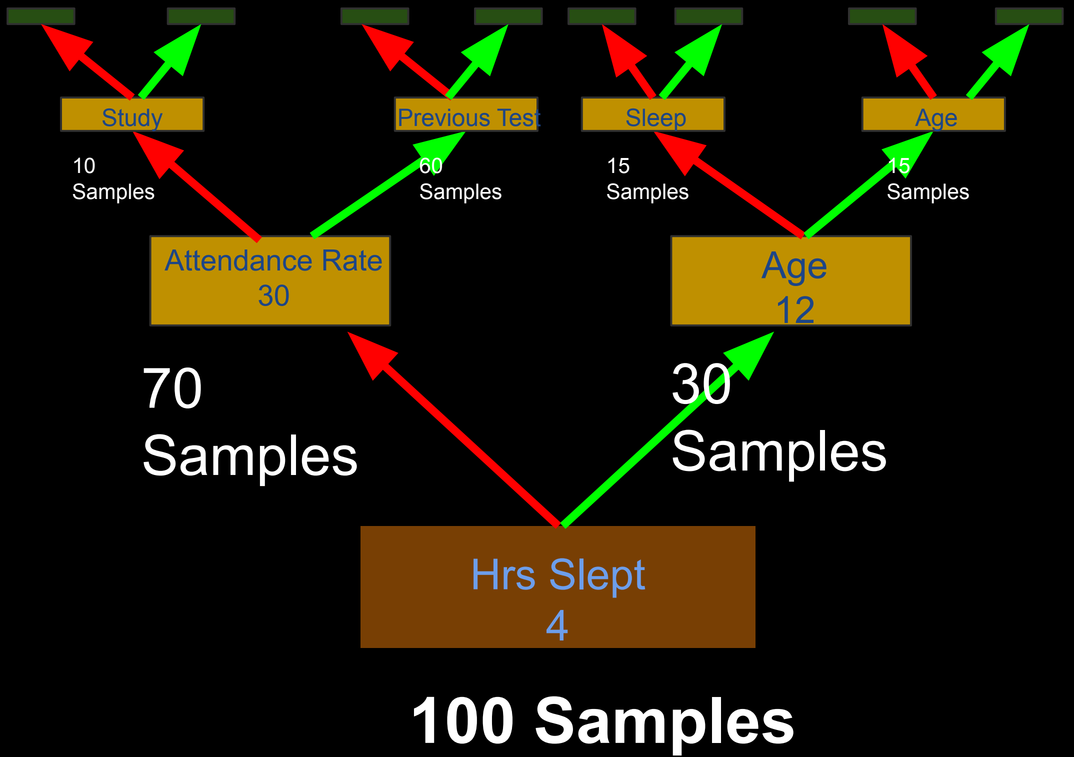
Decision Trees

- **Decision Tree**
- Root node
- Child node
- Left node
- Right node
- **Features**
- **Samples**
- **Target Variable**



Decision Trees

- **Decision Tree**
- Root node
- Child node
- Left node
- Right node
- **Features**
- **Samples**
- **Target Variable**



Experiment

100 Students took a test

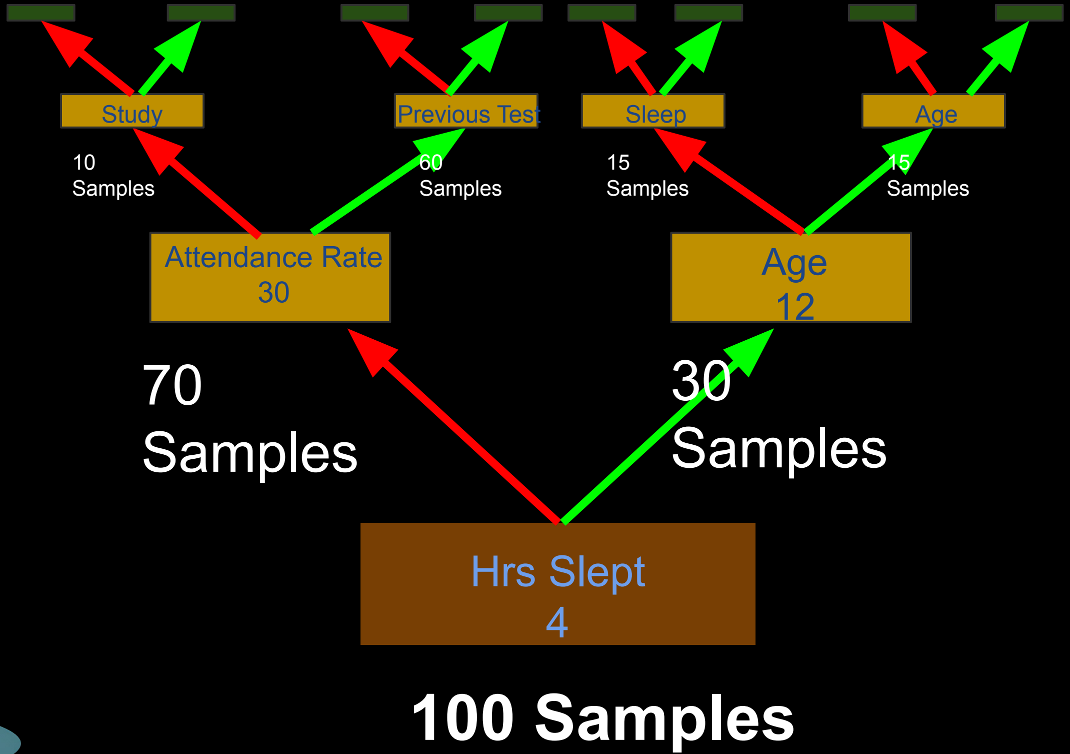
<u>What we know:</u>

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Pass/Fail**

<u>What we want to know:</u>

1. Which factor contributes most to whether a student passes?
2. If another student took the test, based on the prior knowledge, will they pass or fail?

Decision Trees



Experiment

100 Students took a test

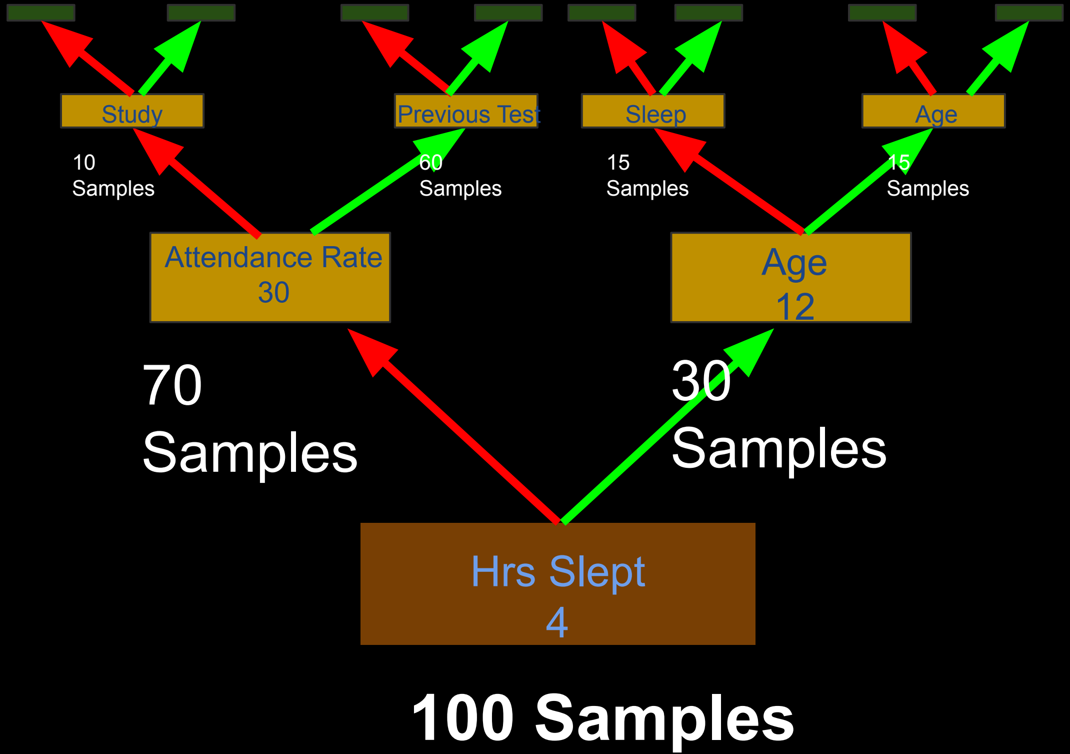
What we know:

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Pass/Fail**

What we want to know:

1. Which factor contributes most to whether a student passes?
2. If another student took the test, based on the prior knowledge, will they pass or fail?

Decision Trees



<u>Feature Importance</u>

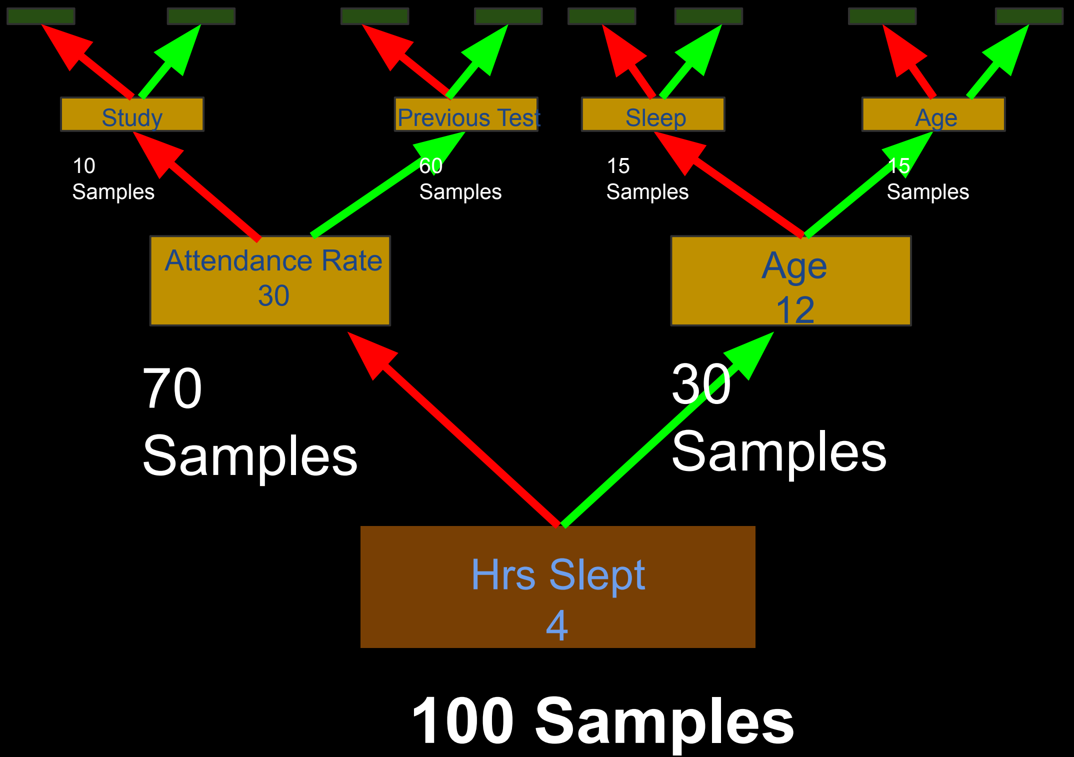
How well does it split when used to split?

Random Forest

- One tree has bias
- It is normal to use a forest

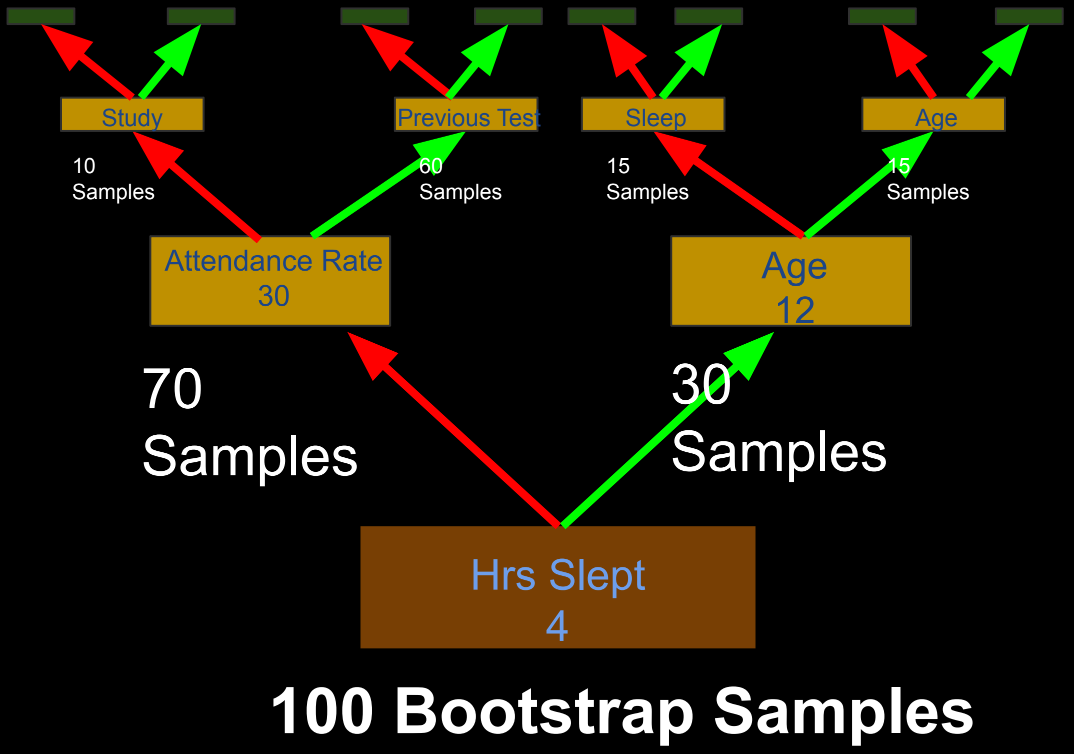
Random Forest

- One tree has bias
- It is normal to use a forest

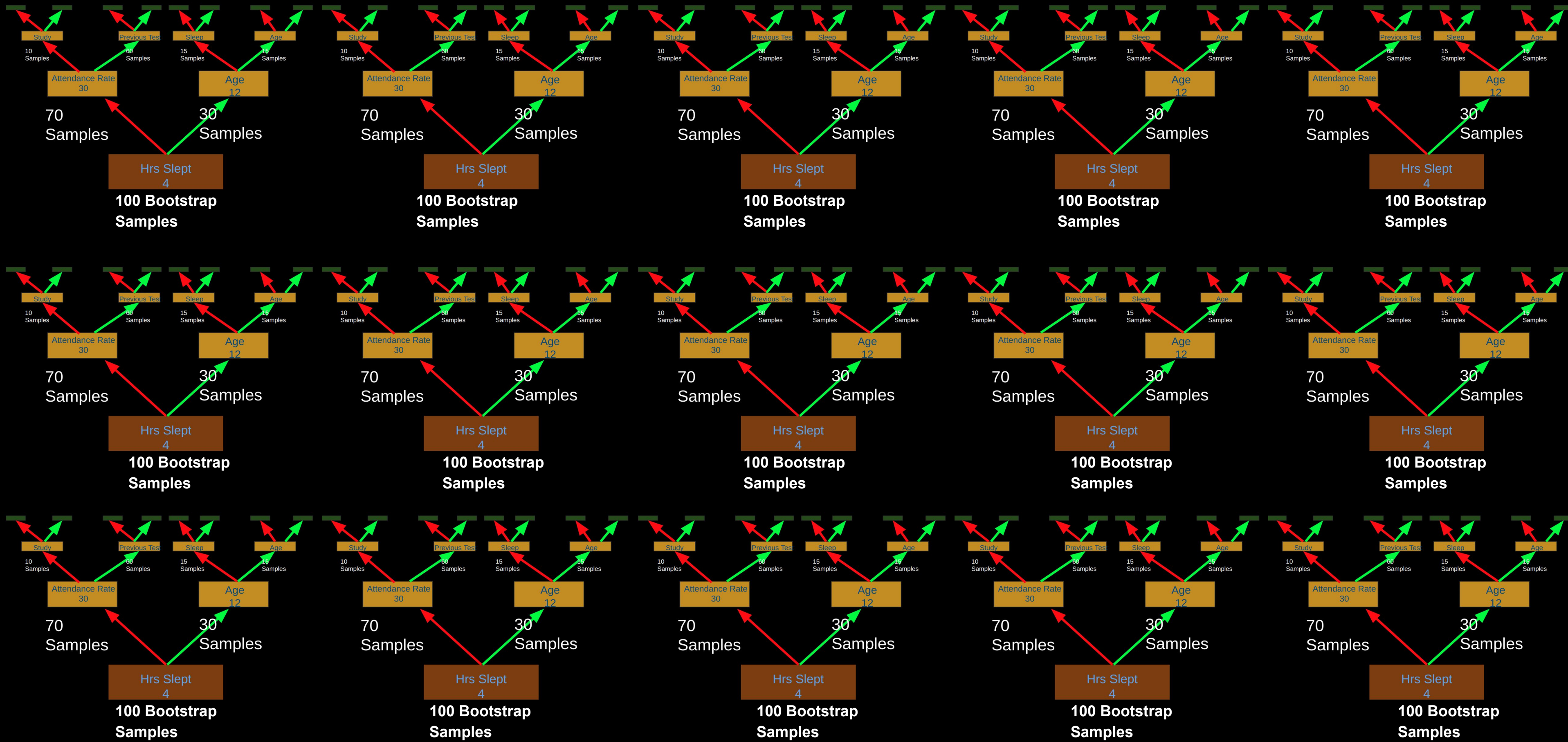


Random Forest

- Lets use 15 Trees
- Bootstrap Sample
  - Size still 100
  - Some values duplicated
  - Some values missing



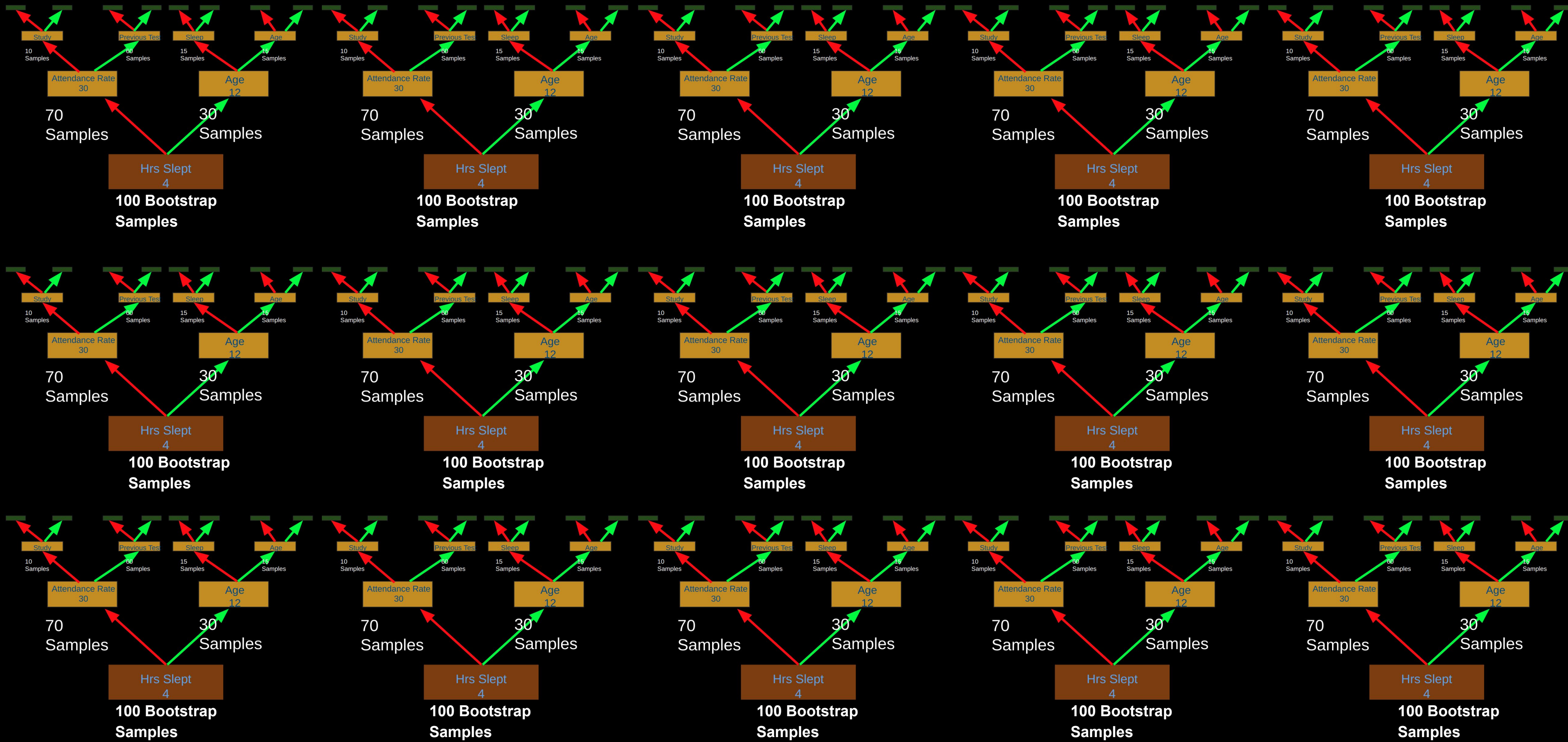
Random Forest



Random Forest

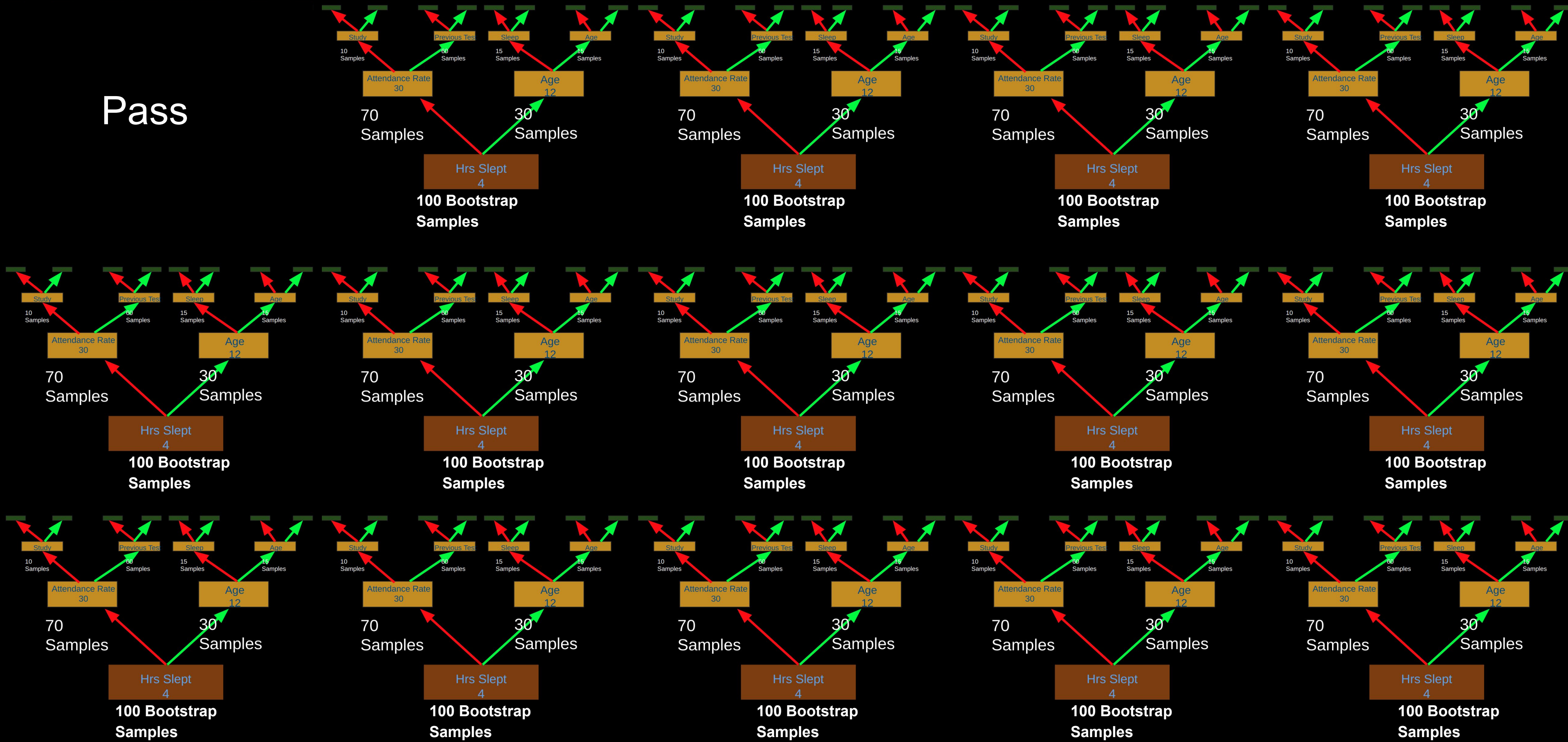
Will they Pass or Fail?

Random Forest



Random Forest

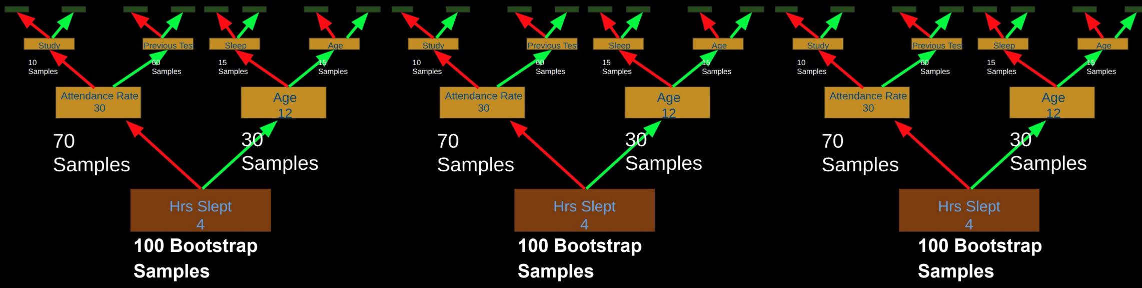
Pass

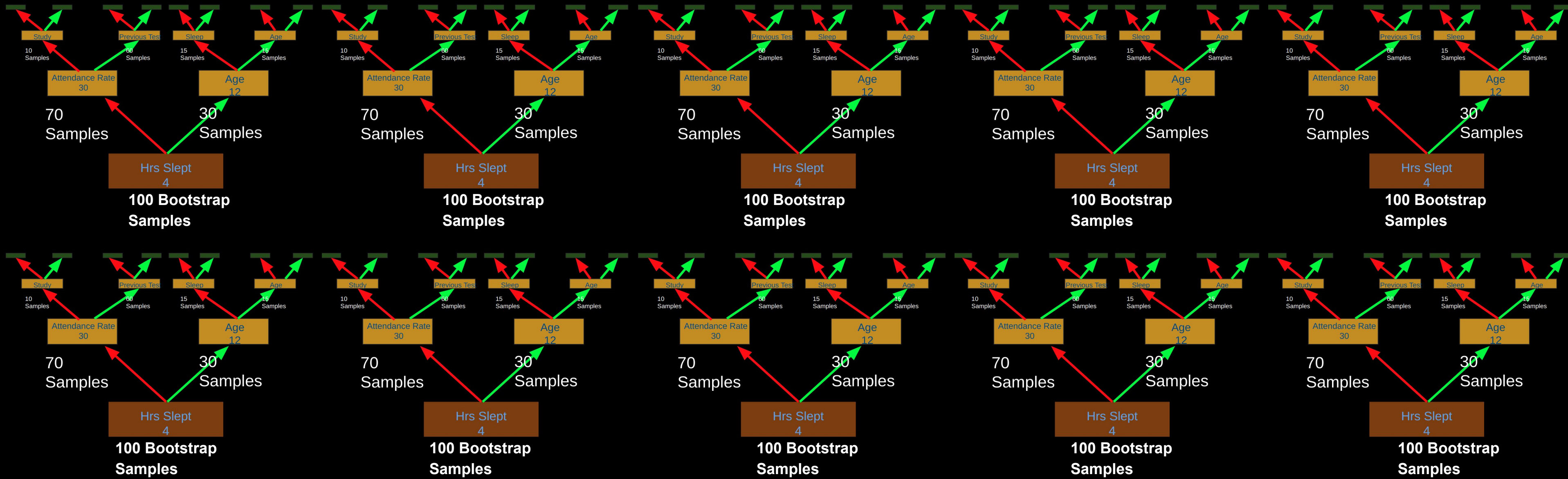


Random Forest

Pass

Fail



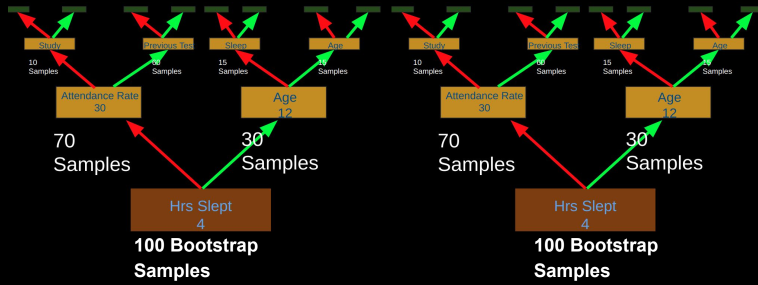


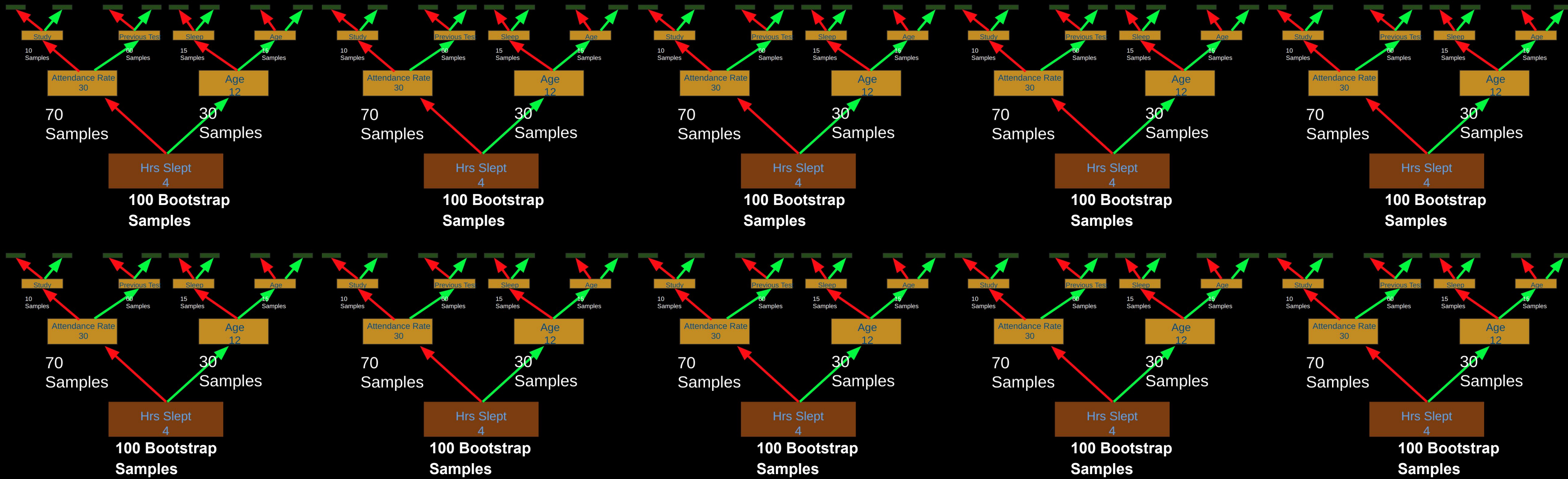
Random Forest

Pass

Fail

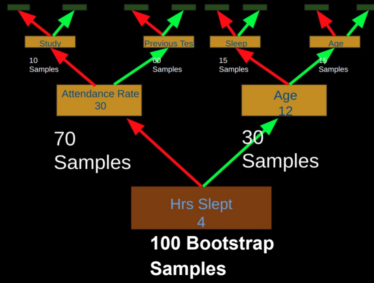
Pass

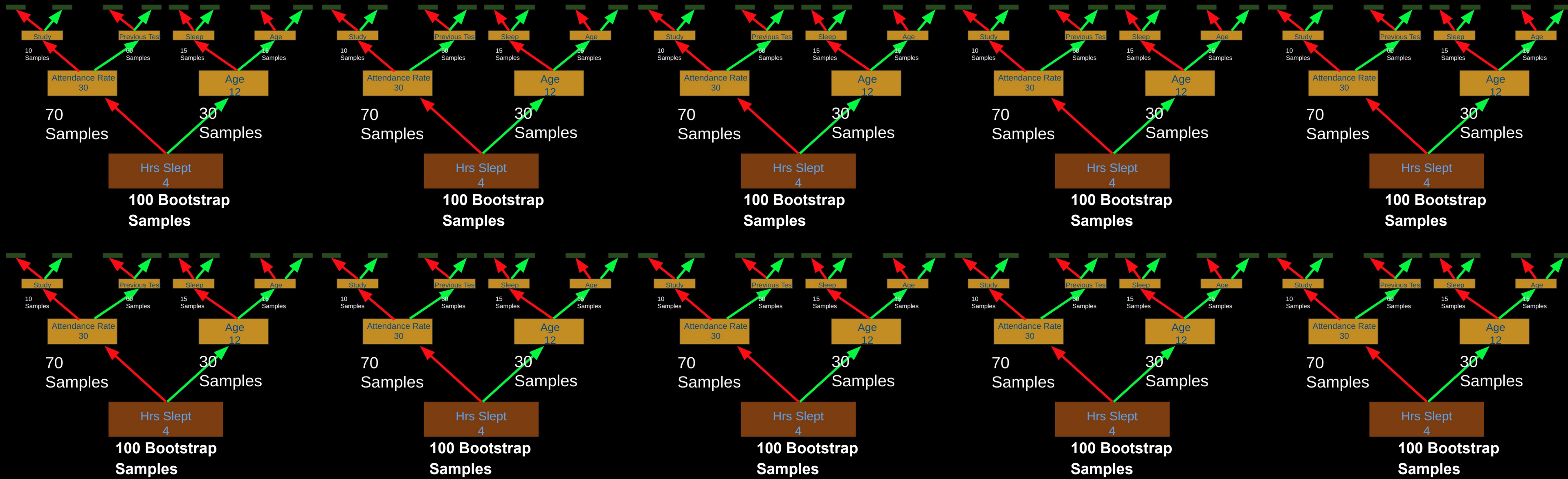




Random Forest

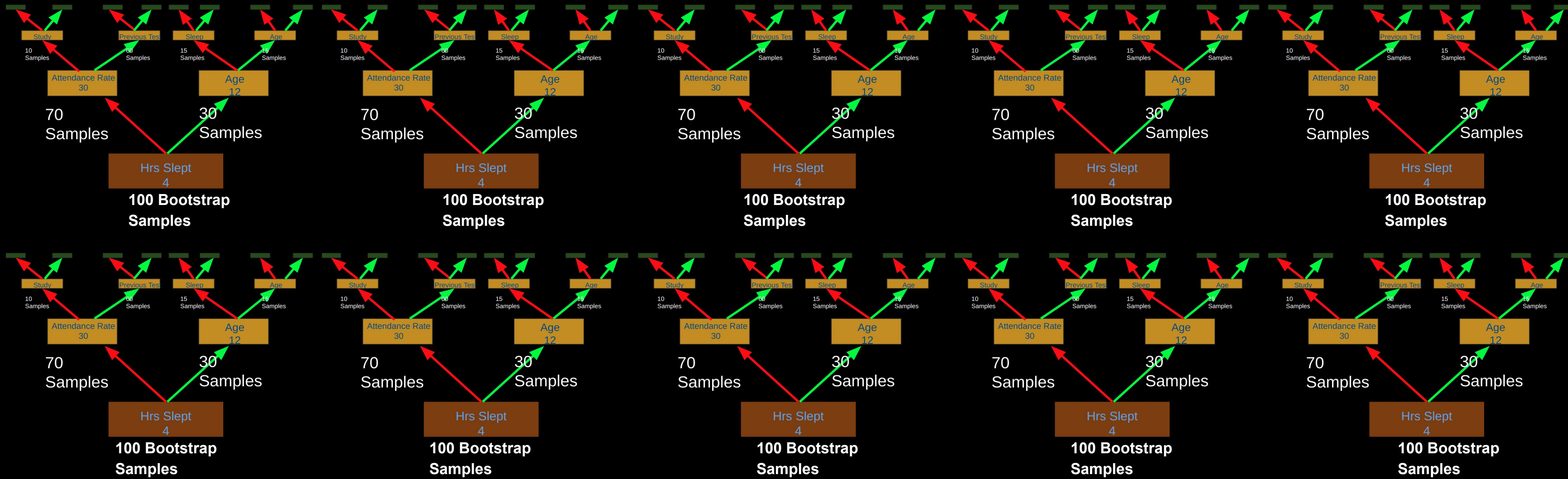
Pass Fail Pass Pass





Random Forest

Pass Fail Pass Pass Pass



Random Forest

Pass	Fail	Pass	Pass	Pass
Pass	Pass	Pass	Pass	Pass
Pass	Pass	Pass	Fail	Pass

Random Forest

Pass	Fail	Pass	Pass	Pass
Pass	Pass	Pass	Pass	Pass
Pass	Pass	Pass	Fail	Pass

Experiment

100 Students took a test

What we know:

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Pass/Fail**

What we want to know:

1. Which factor contributes most to whether a student passes?
2. If another student took the test, based on the prior knowledge, will they pass or fail?

Experiment

100 Students took a test

<u>What we know:</u>

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Pass/Fail**

<u>Binary Target Variable</u>
<u>i.e. True or False</u>

For a binary target variable we can use a ***<u>Random Forest Classifier</u>***

Experiment

100 Students took a test

<u>What we know:</u>

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Pass/Fail**

<u>What we want to know:</u>

1. Which factor contributes most to whether a student passes?

2. If another student took the test, based on the prior knowledge, will they pass or fail?

Experiment

100 Students took a test

What we know:

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Grade**

What we want to know:

1. Which factor contributes most to a students grade?
2. If another student took the test, based on the prior knowledge, what will their grade be?

Decision Tree

- **5 Features** **Hrs studied, Hrs slept, Attendance Rate, Previous Exam Score, Age**
- **Target Variable** **Grade**
- **100 Samples** **Students**

Decision Tree

Root Node

100 Samples

<u>Root Node</u>

- Randomly Select a subset of **Features**

Decision Tree

Hrs studied, Hrs slept, Attendance Rate, Previous Exam Score, Age

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Tree

Hrs studied, **Hrs slept**, **Attendance Rate**, Previous Exam Score, **Age**

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Tree

<u>Root Node</u>

- Randomly Select a subset of **Features**

Loop through **<u>Hrs slept</u>**, **<u>Attendance Rate</u>**, **<u>Age</u>**

Root Node

100 Samples

Decision Tree

Hrs slept (x)

[1, 10, 6, 3 …]

Test each unique value as a threshold

Root Node

100 Samples

Root Node

- Randomly Select a subset of **Features**

Decision Tree

Hrs slept (x)

[1, 10, 6, 3 …]

Threshold

x<1, x≥=1

Left | Right

How well split is the grade?

Root Node

100 Samples

Variance of left and right quantifies the goodness of the split

Root Node

- Randomly Select a subset of **Features**

Decision Trees

<u>Hrs slept</u> (x)

[1, 10, 6, 3 …]

Threshold

x<10, x≥=10

Left | Right

<u>How well split is the grade?</u>

<u>Root Node</u>

- Randomly Select a subset of **Features**

Root Node

100 Samples

Variance of left and right quantifies the goodness of the split

Decision Trees

Hrs slept (x)

[1, 10, 6, 3 …]

Threshold

$x<6$, $x\geq=6$

Left | Right

How well split is the grade?

Root Node

100 Samples

Variance of left and right quantifies the goodness of the split

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Hrs slept (x)

[1, 10, 6, 3 …]

Threshold

$x<3$, $x\geq=3$

Left | Right

How well split is the grade?

Root Node

100 Samples

Variance of left and right quantifies the goodness of the split

Root Node

- Randomly Select a subset of **Features**

Decision Trees

<u>Root Node</u>

- Randomly Select a subset of **Features**

<u>**Hrs slept**</u> (x)

[1, 10, 6, 3 …]

Threshold

x<..., x≥=...

Left | Right

<u>How well split is the grade?</u>

Root Node

100 Samples

Variance of left and right quantifies the goodness of the split

Decision Trees

Attendance Rate (x)

[1, 10, 6, 3 …]

Threshold

x<..., x≥=...

Left | Right

How well split is the grade?

Root Node

100 Samples

Variance of left and right quantifies the goodness of the split

Root Node

- Randomly Select a subset of **Features**

Decision Trees

Age (x)

[1, 10, 6, 3 …]

Threshold

x<..., x≥=...

Left | Right

How well split is the grade?

Root Node

100 Samples

Variance of left and right quantifies the goodness of the split

Root Node

- Randomly Select a subset of **Features**

Decision Trees

List of Impurity
Reduction Values

Which one is best?

Hrs Slept

Hrs Slept
4

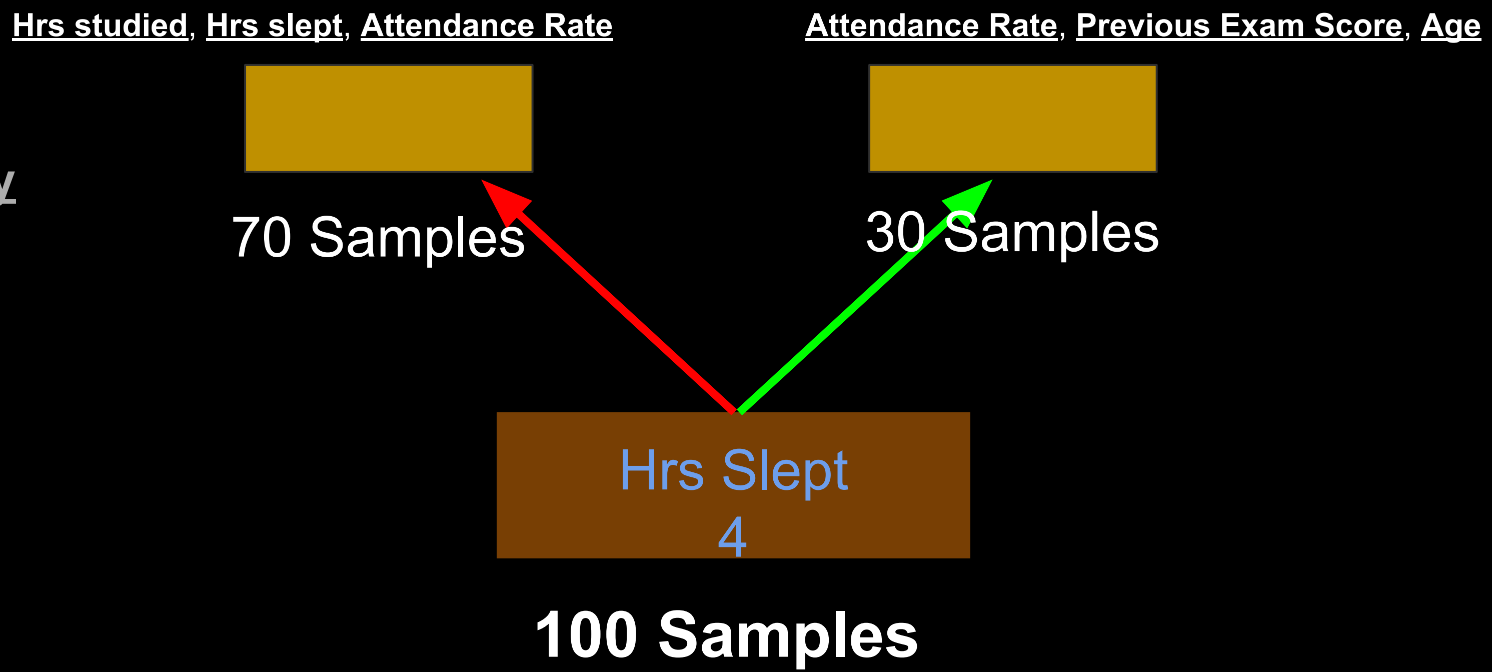
100 Samples

Root Node

- Randomly Select a subset of **Features**

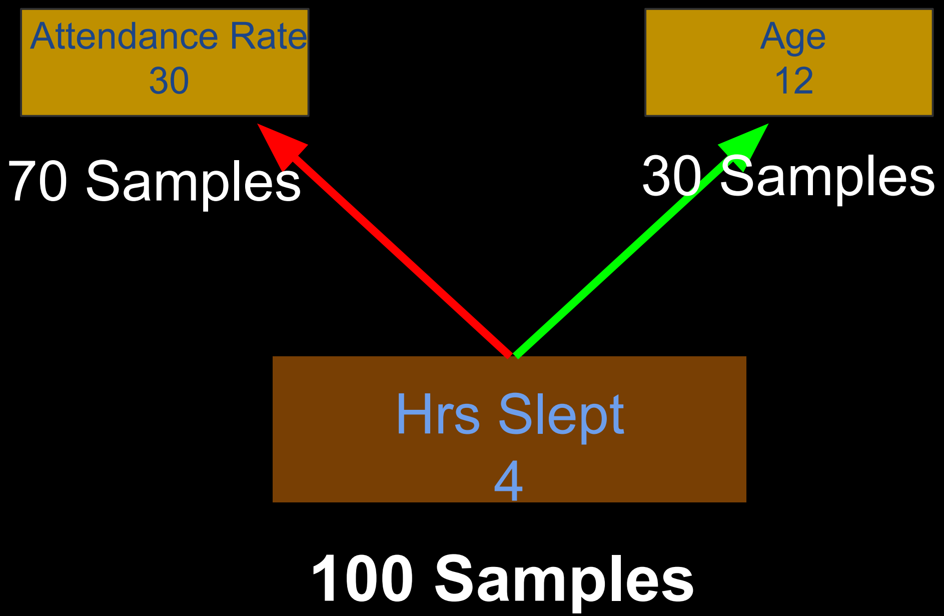
Decision Trees

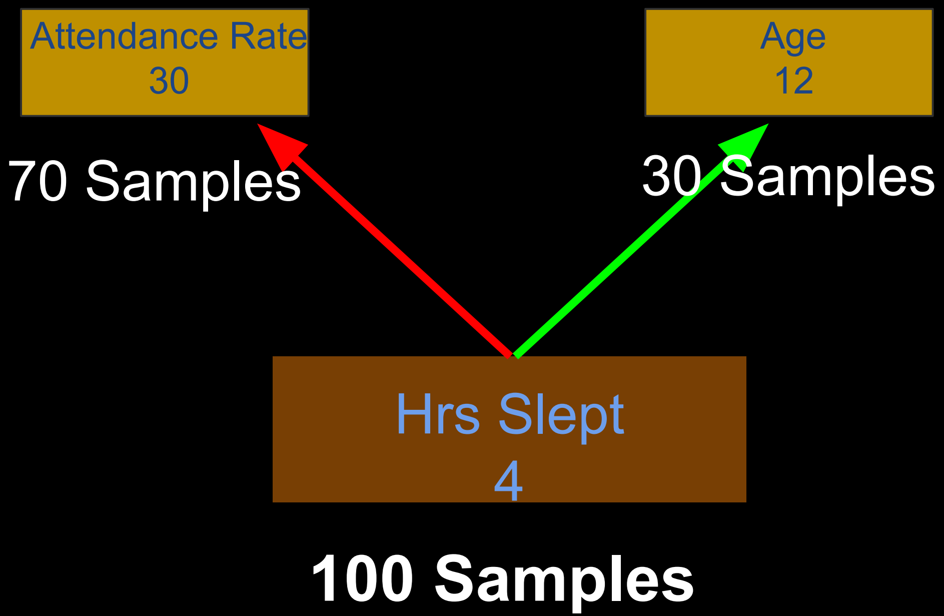
Decision Trees

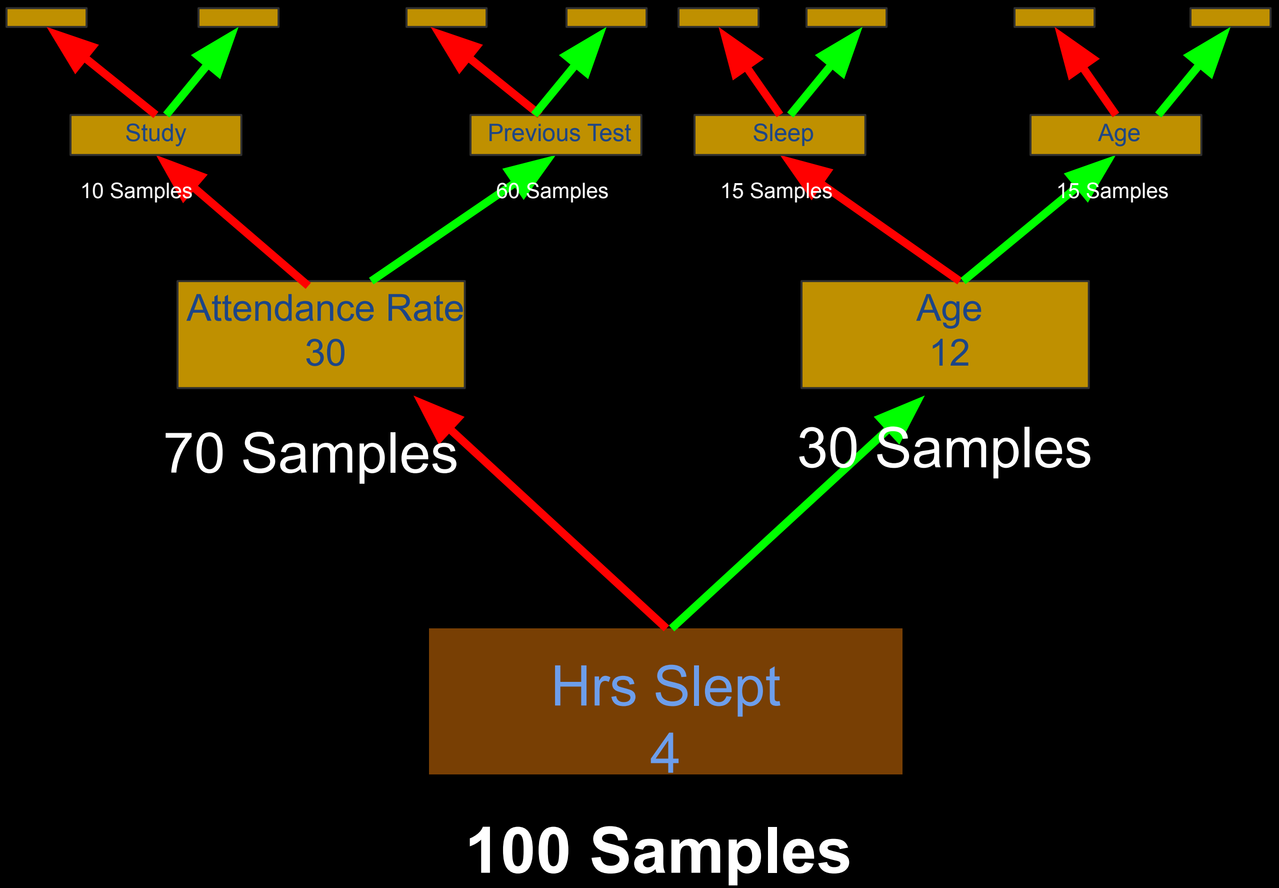


Randomly Select Features

Decision Trees




Attendance Rate
30
Age
12
70 Samples
30 Samples
Hrs Slept
4
100 Samples


Study
Previous Test
Sleep
Age
10 Samples
60 Samples
15 Samples
15 Samples
Attendance Rate
30
Age
12
70 Samples
30 Samples
Hrs Slept
4
100 Samples

Experiment

100 Students took a test

What we know:

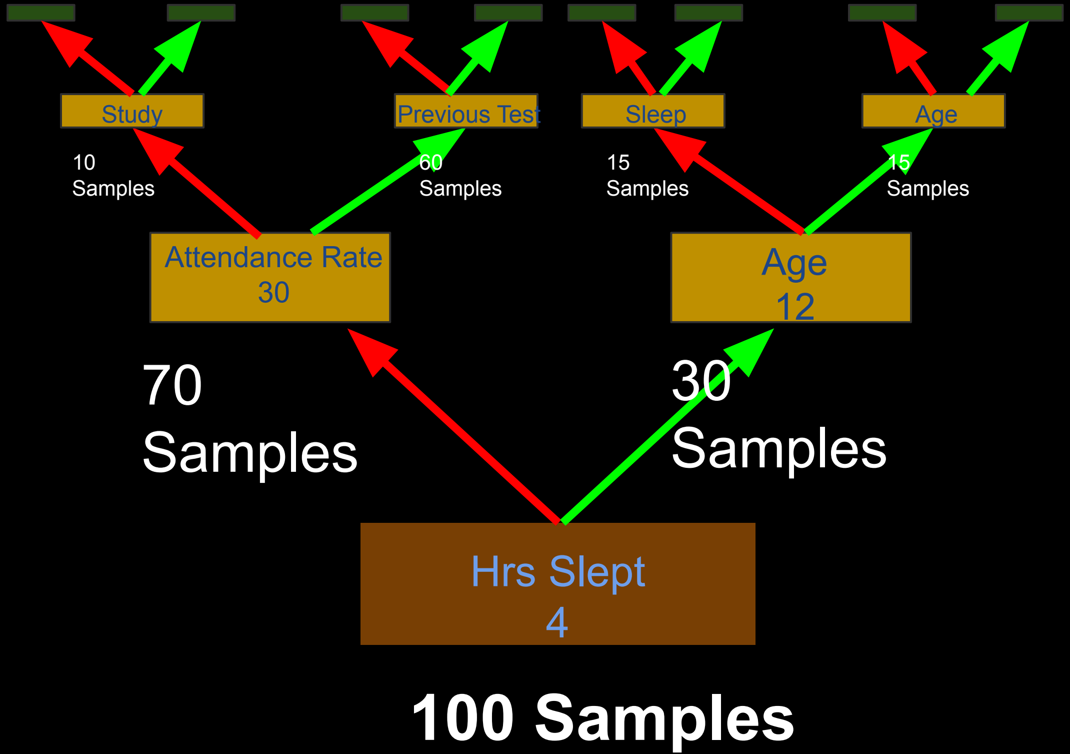
- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Grade**

What we want to know:

1. Which factor contributes most to a students grade?

2. If another student took the test, based on the prior knowledge, what will their grade be?

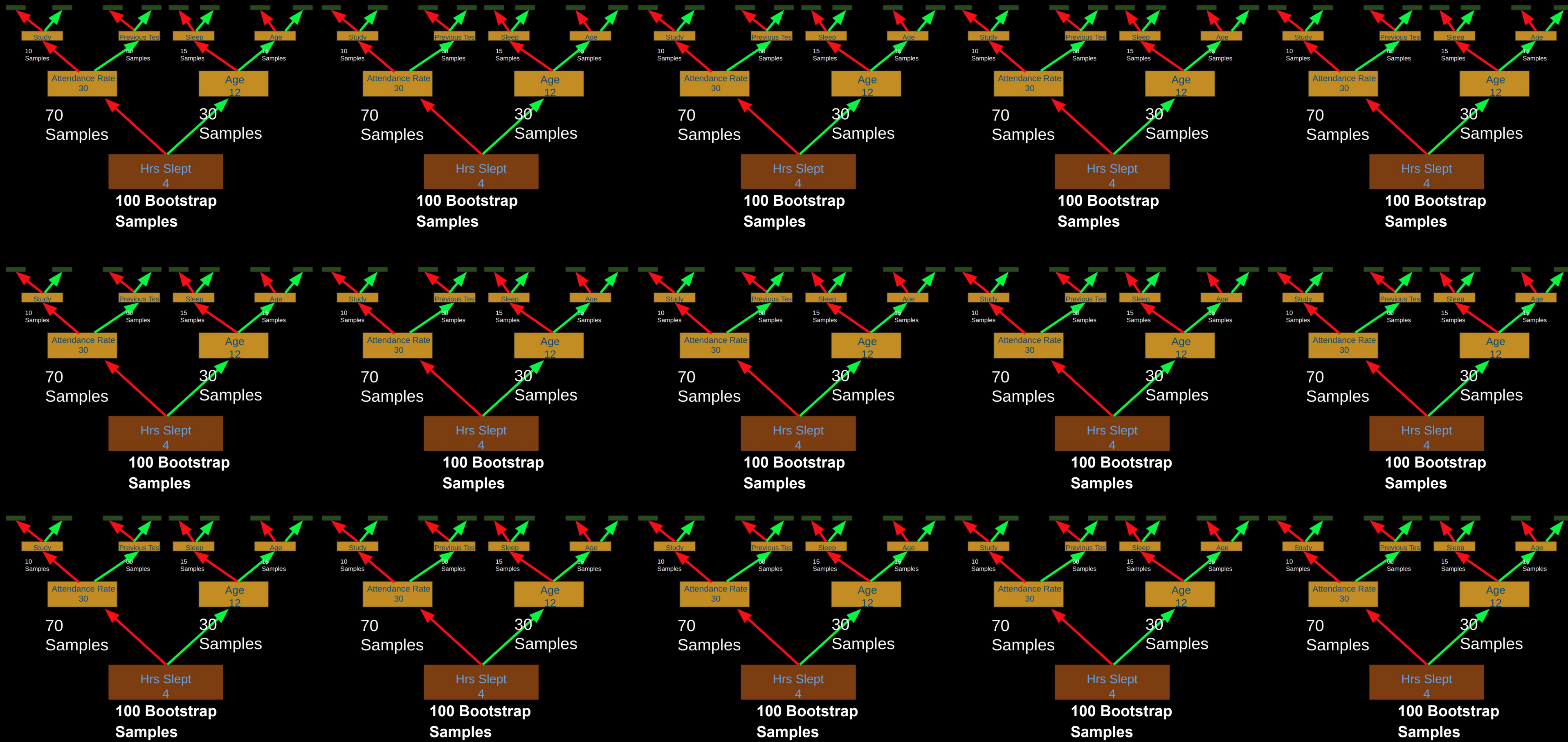
Decision Trees



Decision Trees

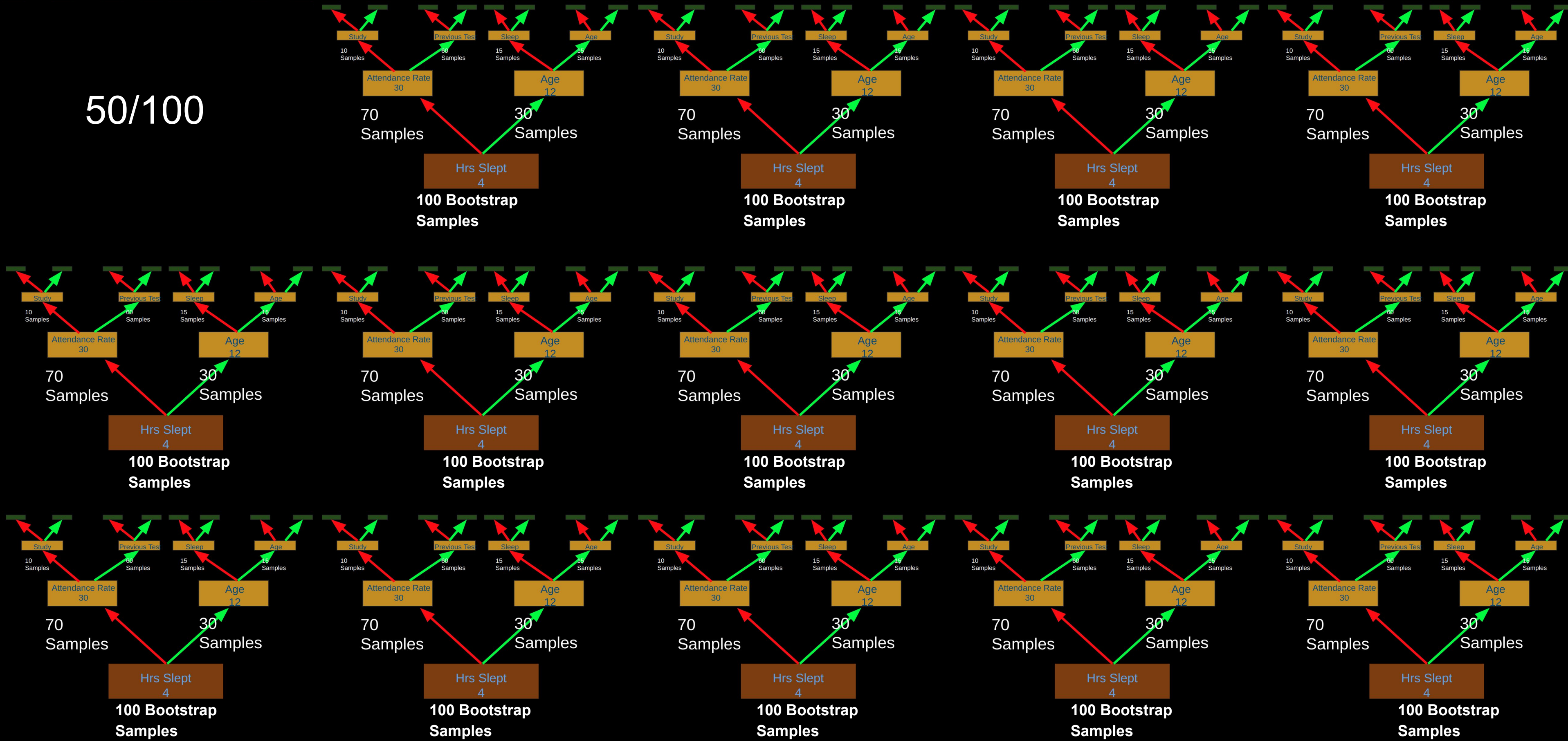
70/100

Random Forest



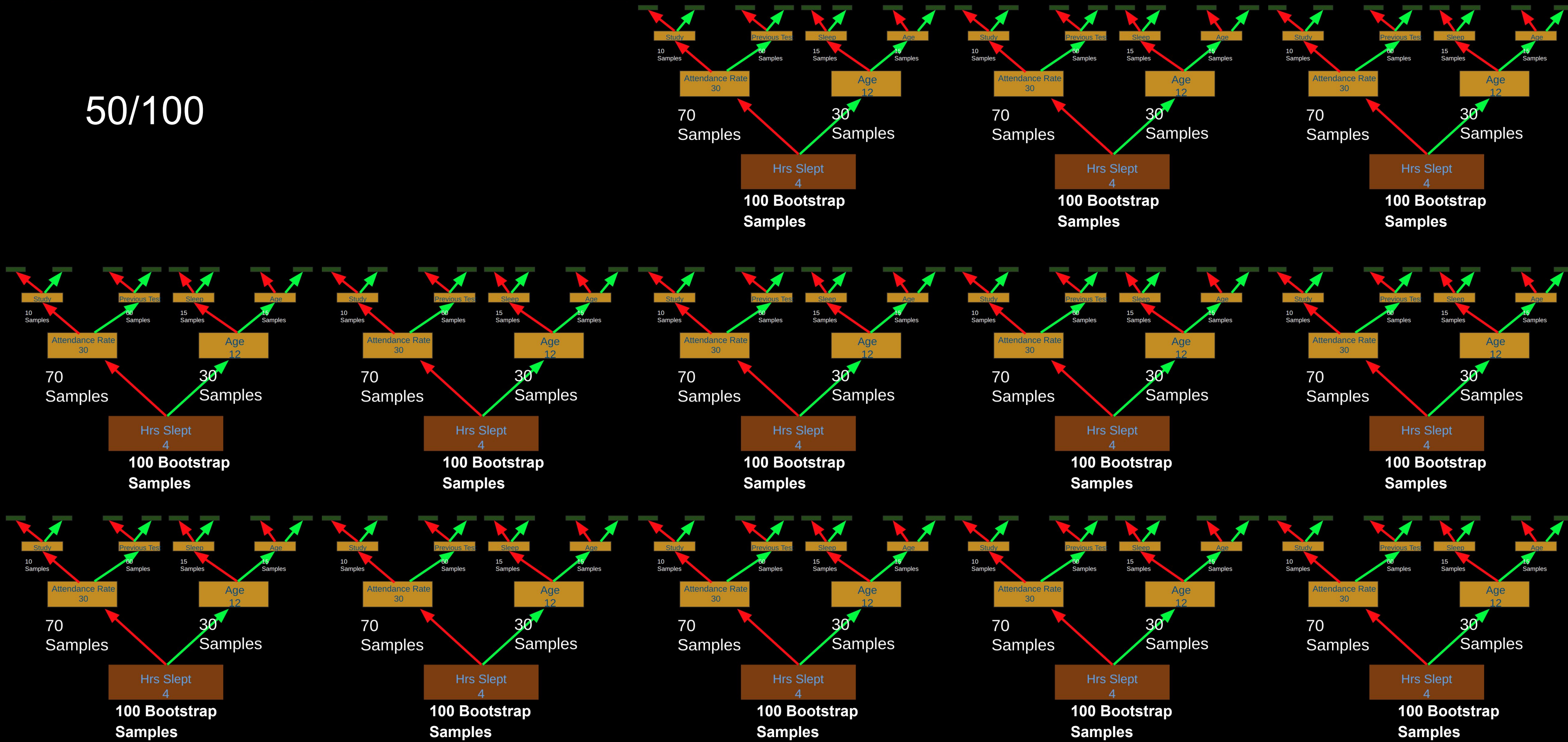
Random Forest

50/100



Random Forest

50/100

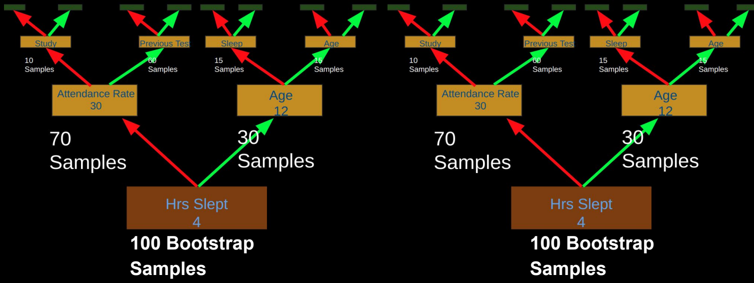


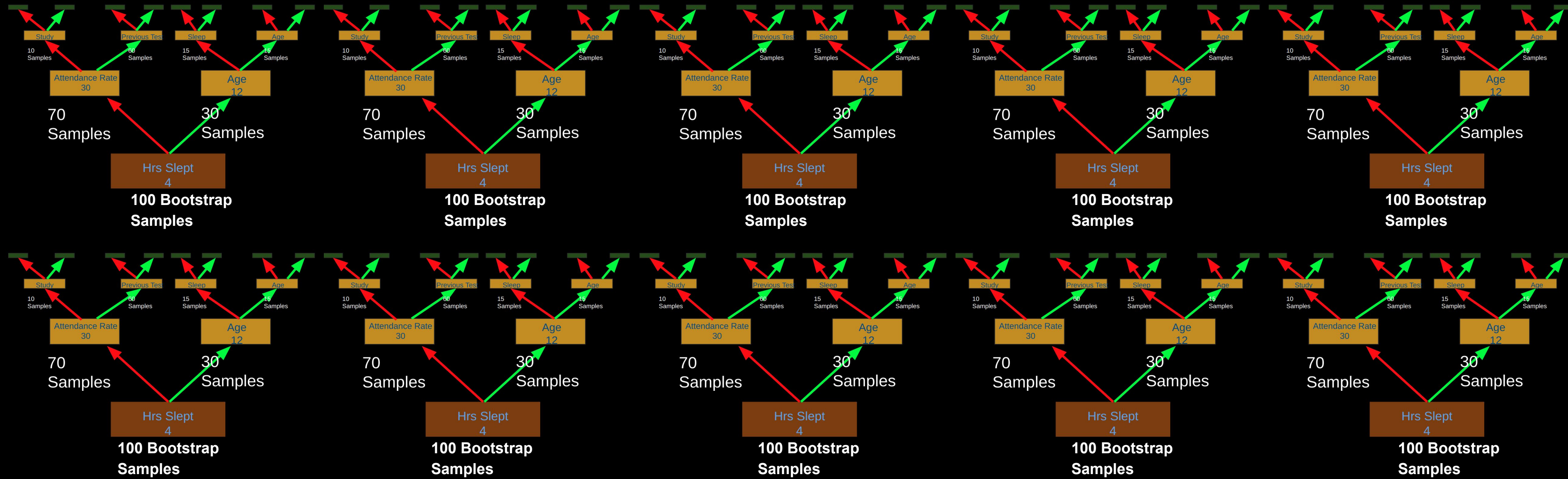
Random Forest

50/100

90/100

20/100





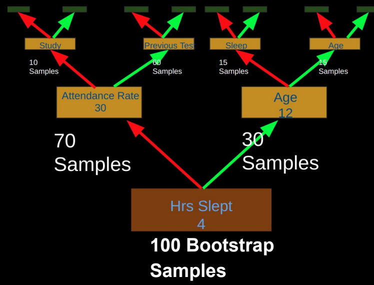
Random Forest

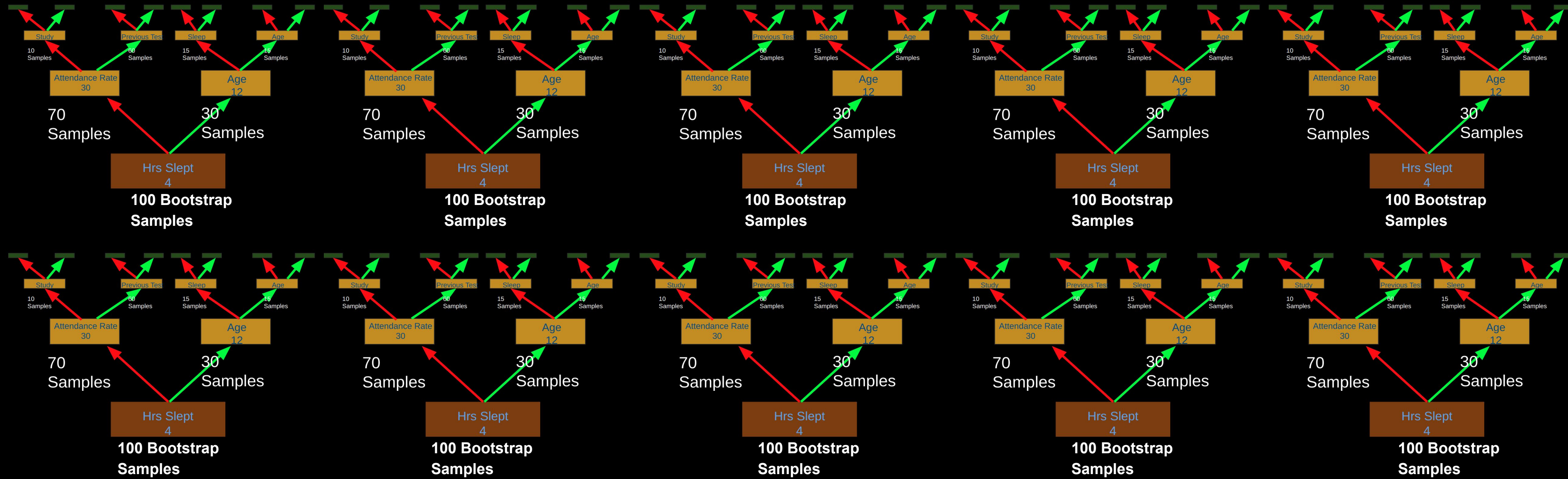
50/100

90/100

20/100

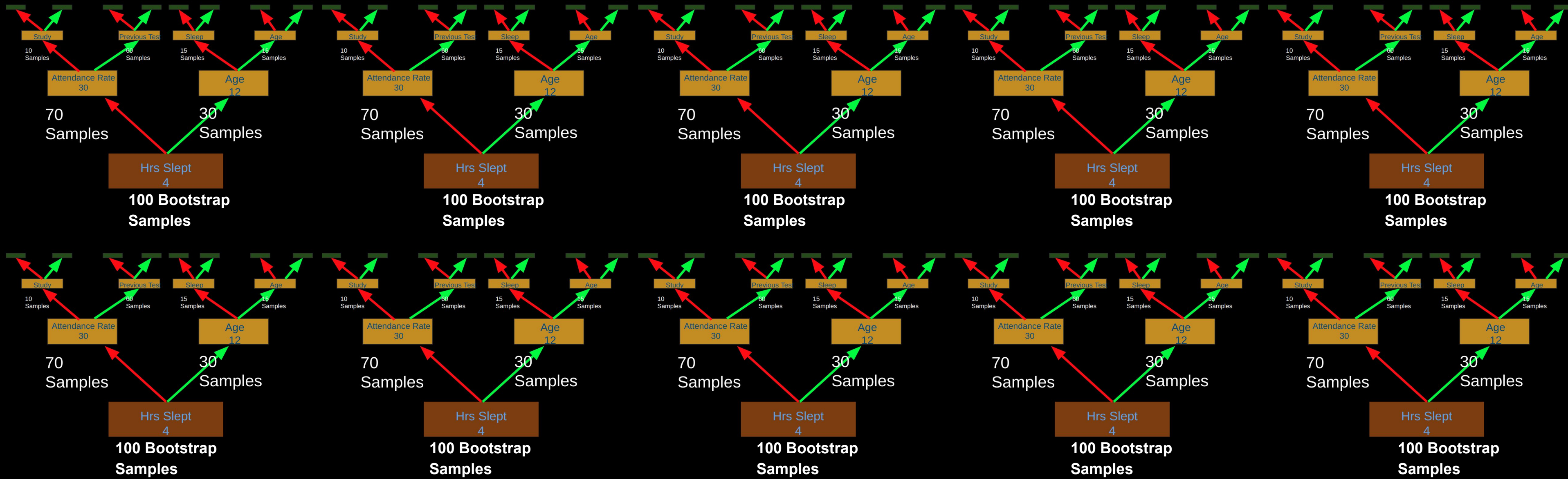
30/100





Random Forest

50/100 90/100 20/100 30/100 51/100



Random Forest

50/100	90/100	20/100	30/100	51/100
55/100	63/100	58/100	70/100	60/100
64/100	62/100	65/100	99/100	58/100

Decision Trees

[50, 90, 20, 30, 51, 55, 63, 58, 70, 60, 64, 62, 65, 99, 58]

Decision Trees

Mean: 60/100

Experiment

100 Students took a test

What we know:

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Grade**

What we want to know:

1. Which factor contributes most to a students grade?
2. If another student took the test, based on the prior knowledge, what will their grade be?

Decision Trees

<u>Feature Importance</u>

How well does it split when used to split?

Experiment

100 Students took a test

What we know:

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Grade**

What we want to know:

1. Which factor contributes most to a students grade?

2. If another student took the test, based on the prior knowledge, what will their grade be?

Experiment

100 Students took a test

What we know:

- **Hrs studied**
- **Hrs slept**
- **Attendance Rate**
- **Previous Exam Score**
- **Age**
- **Grade**

Continuous Target Variable i.e. True or False

For a continuous target variable we can use a ***Random Forest Regressor***

Let's do some coding!